What are large la (LLMs)?

Machine learning

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LLM customization
LLM alignment
AI image generation
Multimodal AI
Retrieval augmented generation (RAG)

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What are LLMs?

Large language models (LLMs) are a
trained on immense amounts of data
understanding and generating natura
content to perform a wide range of ta
neural network architecture called a
handling sequences of words and ca

LLMs work as giant statistical prediction mach
a sequence. They learn patterns in their text a
patterns.

LLMs represent a major leap in how humans ir
first AI system that can handle unstructured h
communication with machines. Where traditic
programmed systems used algorithms to mat
nuance and reasoning. LLMs, once trained, ca
interpreting text, like summarizing an article,
When given agentic capabilities, LLMs can per
various tasks that would otherwise be perforn

LLMs are the culmination of decades of progre
machine learning research, and their developr
of artificial intelligence advancements across
become household names, bringing generativ
LLMs are also used widely in enterprises, with
numerous business functions and use cases.

LLMs are easily accessible to the public throug
AI's ChatGPT, Microsoft's Copilot, Meta's Llam
along with its BERT and PaLM models. IBM ma
which has become the generative AI backbon
Assistant and watsonx Orchestrate.

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Pretraining large lan

Training starts with a massive amount of data-
articles, websites, code and other text source
processing to remove errors, duplication and

This text is broken down into smaller, machine
process of “tokenization.” Tokens are smaller
This standardizes the language so rare and no

LLMs are initially trained with self-supervised
uses unlabeled data for supervised learning. S
labeled datasets, but it’s closely related to su
performance against a "ground truth." In self-
that ground truth can be inferred from unlabel
“correct output” is for each input, as in superv
patterns, structures or relationships in the dat

How Large Language Models Work (5:33 min)

Self-attention

The model passes the tokens through a transf
introduced in 2017, are useful due to their sel
“pay attention to” different tokens at different
of the transformer and its prime innovation. S
the AI model to calculate the relationships an

ones that are distant from one another in the t
parallelization, making the process much mor
qualities allowed LLMs to handle unprecedent

Once text is split into tokens, each token is ma
embedding. Neural networks consists of layer
performs a mathematical operation. Transforr
each, the embeddings are slightly adjusted, be
from layer to layer.

The goal in this process is for the model to lea
that words like “bark” and “dog” appear close
dogs than “bark” and “tree” would, based on
essay. Transformers also add positional encoc
about its place in the sequence.

To compute attention, each embedding is proj
learned weight matrices: a query, a key, and a
token is “seeking,” the key represents the info
value “returns” the information from each key
weight.

Alignment scores are then computed as the si
scores, once normalized into attention weight
flows into the representation of the current to
focus on relevant context while ignoring less i

Self-attention thus creates “weighted” conne
than earlier architectures could. The model as
the tokens. LLMs can have billions or trillions
parameter, the internal configuration variable
how it processes data and makes predictions.
many of these variables exist in a model, with
So-called small language models are smaller i
parameters, making them suitable for deploy
constrained environments.

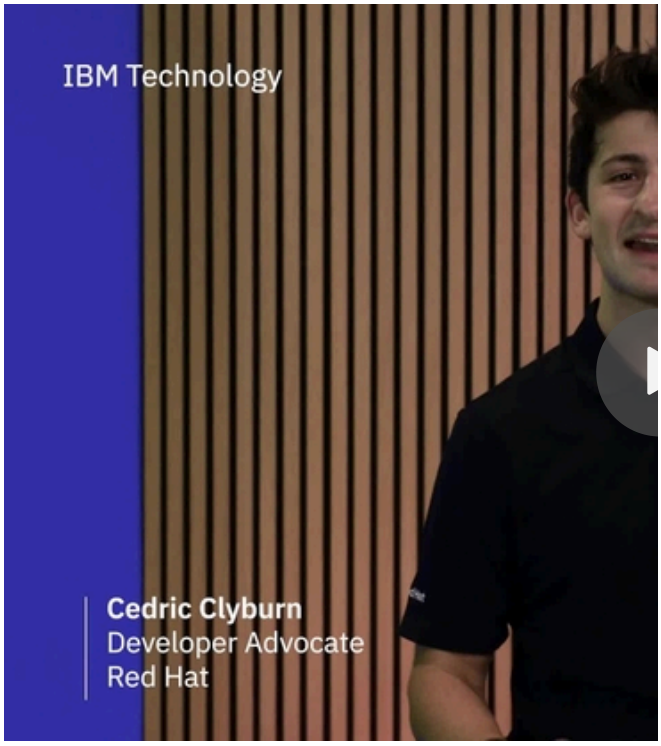
During training, the model makes predictions
training data, and a loss function quantifies th
iterative cycle of making predictions and then
backpropagation and gradient descent, the m
that produce the query, key and value vectors

Once those weights are sufficiently optimized,
vector embedding and produce query, key anc
with the vectors generated for all the other tol
in turn result in attention weights which help t
result is a model that has learned patterns in g
styles and more.

Fine-tuning large lar

After training (or in the context of additional tr
to make them more useful in certain contexts.
on a large dataset of general knowledge can b
order to create a chatbot for the legal field.

Here are some of the most common forms of f
method or a combination of several.



Fine Tuning Large Language Models with InstructL

Supervised fine-tuning

Fine-tuning most often happens in a supervise
dataset. The model updates its weights to bet
labeled data).

While pretraining is intended to give the mode
adapts a general-purpose model to specific ta
customer support. These *functional adaptatio*

fine-tuning produces outputs closer to the hu
resources than training from scratch.

Supervised fine-tuning is also useful for *doma*
model on medical documents so it has the abi

Reinforcement learning f

To further refine models, data scientists often
feedback (RLHF), a form of fine-tuning where
trained to prefer outputs that humans rank hig
process which consists of making LLM outputs
values.

RLHF is also particularly useful for *stylistic alig*
respond in a way that's more casual, humorou
involves training for the same types of tasks, b

Reasoning models

Purely supervised fine-tuning teaches a mode
necessarily encourage better reasoning, which
Such tasks don't always have abundant labele
used in the creation of reasoning models, LLM
problems into smaller steps, often called "rea
output. Increasingly sophisticated means of tr
reasoning and other multi-step decision-maki

Instruction tuning

Another form of LLM customization is instructi
improve a model's ability to follow human inst
dataset consist entirely of tasks that resemble
the outputs demonstrate desirable responses
not inherently optimized for following instruct
tuning is used to better align the model with u

Using large language

Once trained, large language models work by
prompt, converting it into embeddings, and us
at a time, calculating the probabilities for all p
likely one. This process, called inference, is re
model does not “know” the final answer in ad
it learned in training to predict one token at a

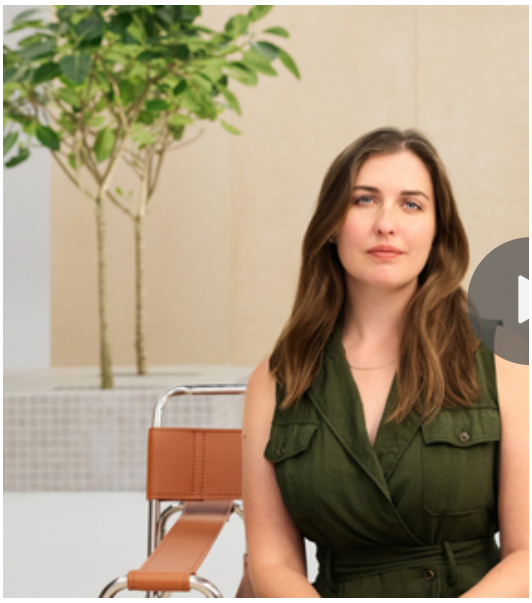
The easiest and fastest way to get domain-spe
is through prompt engineering, which does no
prompts in all sorts of ways. For example, a pr
healthcare professional” could yield more rele
recommended to be used for medical advice!)

LLMs have other strategies to control their out
controls the randomness of text that is genera
sampling, which limits the set of tokens consi
creativity and coherence.

The context window is the maximum number
once when generating text. Early LLMs had sh
of thousands of tokens in their context windov
entire research papers, performing code assis
continuous conversations with users.

Retrieval augmented generation (RAG) is a me
external knowledge bases, enabling them to d
level of accuracy. The retrieved information is
the model can use it when generating respons
by connecting an LLM to a dynamic weather se
information for a user about that day’s weathe

AI Academy



Why foundation models are a

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Deploying LLMs

Building an LLM from scratch is a complex anc
popular LLMs are the result of immense amou
expertise, which is why most are built and ma
expansive resources.

However, many of these models are accessibl
can use pretrained models to build chatbots, l
tools and more. For more control over data an
can be deployed locally or in the cloud. Github
make AI development accessible to all.

Developers can use LLMs as the basis for all k
exciting developments in AI is the agentic sys
themselves, LLMs simply generate text based
memory, APIs, decision logic and other extern
booking a flight or piloting a self-driving vehicl

Large language mod

LLMs are redefining business processes and h
use cases in many industries.

- **Text generation**: LLMs can do all sorts of c
blog posts or legal memos in response to p

- **Text summarization**: LLMs can summarize
corporate documentation and customer hi
a desired output format and style.

- **AI assistants**: Chatbots powered by conve
and provide detailed information as a part
solution.

- **Code generation**: Code assist platforms ai
errors in code and uncovering security issu
translating between them.

- **Sentiment analysis**: Customer tone is ana
feedback at scale.

- **Language translation**: Automated translat
across languages and geographies with flu
- Reasoning: LLMs can solve math problems
complex concepts in simpler terms.

Evaluating LLMs

LLMs are powerful tools, but they come with s
accuracy. During hallucinations, the model ge
while sounding plausible. LLMs can also reflec
data, producing outputs that are unfair or offe
are significant: training and running LLMs requ
and energy, raising both cost and environment

Practitioners can mitigate these negative aspe
governance, the processes, standards and gua
are safe and ethical. A key part of governance
benchmarks. LLM benchmarks provide quanti
models. Because LLMs are general-purpose s
their evaluation requires multiple dimensions

and practitioners look at qualities like accurac
to determine how well a model performs.

LLMs are also evaluated on the basis of alignm
teaming, where evaluators intentionally try to
responses to expose weaknesses. Fairness an
prevent LLMs from reproducing harmful stere

LLMs are also commonly evaluated on the bas
token throughput, memory footprint and abilit
of the common metrics used to evaluate how

A short history of LL

The history of LLMs goes back to the early day
processing, when researchers used rule-base
text. These early approaches could capture lo
long-range dependencies or deeper semantic

A major shift came in the 2010s with the rise
like Word2Vec and GloVe, which represented
enabling models to learn semantic relationshi
neural networks (RNNs) and long short-term r
handle sequential data.

In 2017, Vaswani *et al.* introduced the encode
landmark paper "Attention Is All You Need."[1
models on large datasets, marking the beginn
(2018), an encoder-only transformer, demons
understanding language, while OpenAI's gene
based on a decoder-only variant, showed how
could yield remarkably fluent language genera
decoder models like Google's T5 and Faceboo
sequence-to-sequence design for tasks such
(2019) attracted attention for its ability to gen
(2020), with 175 billion parameters, cemente

In addition, new architectures are challenging
models work by using a state-space model wi
combine past information, allowing it to captu
start with random noise and gradually denoise
until coherent text emerges. Both architecture
transformers.

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Footnotes

1. "Attention is all you need", Vaswani et al, arXiv, 12 June 2017